

Hao Lun Li

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EDUCATION

University of British Columbia, Vancouver

Sep 2025 - Apr 2029

Bachelor of Science in Computer Science, Co-op Program, Dean's List

GPA: 88.3%

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript/JavaScript, C++, HTML/CSS/JSON, R
- **Frameworks & Tools:** React, React Native, Expo, Git, Junit, Tidyverse, GitHub Actions, Rest APIs / OAuth 2.0
- **Cloud & Databases:** AWS (EC2, S3, Aurora & RDS), Supabase (Auth, Tables), PostgreSQL

PERSONAL PROJECTS

Brezi | Typescript, HTML, CSS, React Native, Expo, PostgreSQL

Apr 2026 - Present

- Built a **React Native** photo wellness app with AI judging, designed basic Postgres schema in Supabase with RLS.
- Integrated EfficientDet local real-time AI object detection with React Native Vision Camera and migrated local Windows toolchain to EAS Build, establishing a reliable build-and-iterate workflow.
- In development of a 5-stage submission pipeline with per-user-scoped **cloud** storage on Supabase and client-side image compression via React Native Image Resizer to minimize upload payloads.

calcli (Google Calendar CLI) | C++, CMake, vcpkg, Catch2 | [GitHub](#)

Sep 2026 – Present

- Built a C++ command-line tool that creates Google Calendar events from natural-language input (e.g. calcli add "Exam" Friday 2:30 1 hour), with a unit-tested parser for weekdays, clock times, and durations.
- Implemented the full OAuth 2.0 flow (consent, token exchange, caching, refresh) and a **REST** client over libcurl for the Google Calendar API, with RFC 3339 JSON serialization via nlohmann::json.
- Shipped v0.1.0-alpha as a standalone statically linked Windows binary with a **CI/CD pipeline** using GitHub Actions that runs the full **Catch2** test suite and publishes releases on version tags.

TECHNICAL EXTRACURRICULAR ACTIVITIES

Team SHIJ, Amazon Robotics Hackathon 2026 | Git, Python | [GitHub](#)

Sep 2026

- Designed and implemented a Dijkstra-based dynamic routing algorithm in Python for warehouse drive units in a team of 4, caching reverse-graph shortest-path distances so units re-evaluate the optimal next hop every timestep.
- Built congestion and deadlock-aware coordination logic to prevent race conditions between idle units as well as integrated detour-vs-wait cost comparisons using live data to achieve **100%** pod delivery (vs. 66.7% for a baseline greedy algorithm) and completing all test scenarios in **~75%** fewer time steps.

Generic Health Consulting, WiDS ASA Datafest 2026 | R, Python | [GitHub](#)

Mar 2026

- Analyzed over **340 000** lines of distributed medical history information from Stormont Vail Health in **R** to accurately locate and evaluate misallocated post-operative attention on varying patient needs.
- Classified patients into 4 archetypes through **KNN-Classification** providing Stormont Vail Health with better triage tools improving patient recovery and lowering hospital operating costs via more effective prioritization.

WORK EXPERIENCE

Aquatic Leader, City of Coquitlam

Feb 2024 - Present

- Communicated with and lifeguarded up to **500** patrons daily, addressing concerns, answering questions, and resolving issues efficiently in a fast-paced environment.